

Last login: Wed Jun 17 16:04:07 on ttys002

```
psql
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
zsh: command not found: psql
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.
```

```
tpch=# SET ROLE alice;
SET
tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT count(*)
FROM lineitem;
WARNING:  Log 1: Planner Input:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{SUBLINK
:sublinkType 0
:sublinkid 0
:ctestexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{FUNCEXP
:funcid 17005
:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
:location -1
}
)
:mergeActionList <>
```

```

:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
      }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
}
)
:rtperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals <>
  }
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {AGGREF
        :aggfnoid 2803
        :aggtype 20
        :aggcollid 0
        :inputcollid 0
        :aggtranstype 0
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :agppresorted false
        :agglevelsup 0
        :aggssplit 0
        :aggno -1
        :aggtransno -1
        :location 43
      }
    :resno 1
    :resname count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: max_parallel_hazard_walker: FuncExpr funcid=17005 propparallel=s (s=safe r=restricted u=unsafe)
 WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
}

```

```

:cteList <>
:rtable (
  {RANGETBLENTTRY
    :alias
      {ALIAS
        :aliasname s
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname s
        :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtekind 0
    :reloid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :reloid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals (
      {FUNCEXP
        :funcid 17005
        :funcresulttype 16
        :funcresultset false
        :funcvariadic false
        :funcformat 0
        :funccollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 6
            :vartype 701
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 6
            :location -1
          }
        )
        :location -1
      }
    )
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDepts <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {FUNCEXP :funcid 17005 :funcresulttype 16 :funcresultset false :funcvariadic false :funcformat 0 :funccollid 0 :inputcollid 0 :args ({VAR :v
arno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) :location -1} :is_pushed_down true :can_join f
alse :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :oute
r_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :ou
ter_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: FuncExpr funcid=17005 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {FUNCEXP :funcid 17005 :funcresulttype 16 :funcresultset false :funcvariadic false :funcformat 0 :funccollid 0 :inputcollid 0 :args ({VAR :
varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) :location -1} :is_pushed_down true :can_join
false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :ou
ter_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :ou
ter_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL:
  {PATH
    :pathtype 339
    :parent_relds (b 1)
    :pathtarget
      {PATHTARGET
        :exprs <>
        :sortgrouprefs ()
        :cost.startup 0
        :cost.per_tuple 0
        :width 0
        :has_volatile_expr 0
      }
    :required_outer (b)
    :parallel_aware false
    :parallel_safe true
    :parallel_workers 0
    :rows 333
    :disabled_nodes 0
    :startup_cost 0
    :total_cost 2818
    :pathkeys <>
  }

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
  }

```

```

:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
      }
    :rtekind 0
    :relid 16566
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {PARAM
        :paramkind 1
        :paramid 0
        :paramtype 16
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals <>
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
    :expr
      {AGGREF
        :aggnoid 2803
        :aggtype 20
        :aggcollid 0
        :inputcollid 0
        :aggtranstype 0
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        :agglevelsup 0
        :aggsplit 0
        :aggno -1
        :aggtransno -1
        :location 43
      }
    :resno 1
    :resname count
    :resortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudoconstants false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relids (b) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=0 in_safe_list=1 > reason=outer reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relids (b) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({AGGREF :aggnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREF :aggnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

```

WARNING: Log 3: At the starting of Plan creation:

```
DETAIL: {AGGPATH
: path.pathtype 365
: parent_relids (b)
: required_outer (b)
: path.parallel_aware false
: path.parallel_safe false
: path.parallel_workers 0
: path.rows 1
: path.disabled_nodes 0
: path.startup_cost 133850.37381788183
: path.total_cost 133850.38381788184
: path.pathkeys <>
: subpath
{GATHERPATH
: path.pathtype 368
: parent_relids (b)
: required_outer (b)
: path.parallel_aware false
: path.parallel_safe false
: path.parallel_workers 0
: path.rows 2
: path.disabled_nodes 0
: path.startup_cost 133849.31333333335
: path.total_cost 133849.52333333337
: path.pathkeys <>
: subpath
{AGGPATH
: path.pathtype 365
: parent_relids (b)
: required_outer (b)
: path.parallel_aware false
: path.parallel_safe true
: path.parallel_workers 2
: path.rows 1
: path.disabled_nodes 0
: path.startup_cost 132849.31333333335
: path.total_cost 132849.32333333336
: path.pathkeys <>
: subpath
{PATH
: pathtype 339
: parent_relids (b 1)
: pathtarget
{PATHTARGET
: exprs <>
: sortgrouprefs <>
: cost.startup 0
: cost.per_tuple 0
: width 0
: has_volatile_expr 0
}
: required_outer (b)
: parallel_aware true
: parallel_safe true
: parallel_workers 2
: rows 1250192
: disabled_nodes 0
: startup_cost 0
: total_cost 129723.83333333334
: pathkeys <>
}
: aggstrategy 0
: aggsplit 6
: numGroups 1
: transitionSpace 0
: groupClause <>
: qual <>
}
: single_copy false
: num_workers 2
}
: aggstrategy 0
: aggsplit 9
: numGroups 1
: transitionSpace 0
: groupClause <>
: qual <>
}
```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```
DETAIL: {PLANNEDSTMT
: commandType 1
: queryId 0
: planId 0
: hasReturning false
: hasModifyingCTE false
: canSetTag true
: transientPlan false
: dependsOnRole false
: parallelModeNeeded true
: jitFlags 25
: planTree
{AGG
: plan.disabled_nodes 0
: plan.startup_cost 133850.37381788183
: plan.total_cost 133850.38381788184
: plan.plan_rows 1
: plan.plan_width 0
: plan.parallel_aware false
: plan.parallel_safe false
: plan.async_capable false
: plan.plan_node_id 0
: plan.targetlist {
{TARGETENTRY
: expr
{AGGREF
: aggfnoid 2803
: aggtype 20
: aggcollid 0
: inputcollid 0
: aggtranstype 20
: aggargtypes <>
: aggdirectargs <>
: args {
{TARGETENTRY
: expr
{VAR
: varno -2
: varattno 1
: vartype 20
: vartypmod -1
: varcollid 0
: varnullingrels (b)
: varlevelsup 0
: varreturningtype 0
: varnosyn 0
: varattnosyn 0
: location -1
}
}
: resno 1
: resname <>
: ressortgroupref 0
: resorigtbl 0
: resorigcol 0
: resjunk false
}
}
}
: aggorder <>
: aggdistinct <>
: aggfilter <>
: aggstar true
: aggvariadic false
: aggkind n
: aggpresorted false
: agglevelsup 0
: aggsplit 9
: aggnos 0
}
```

```

:aggtransno 0
:location 43
}
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{GATHER
:plan.disabled_nodes 0
:plan.startup_cost 133849.31333333335
:plan.total_cost 133849.52333333337
:plan.plan_rows 2
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 132849.31333333335
:plan.total_cost 132849.32333333336
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 6
:aggno 0
:aggtransno 0
:location 43
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SESCAN
:scan.plan.startup_cost 0
:scan.plan.total_cost 129723.83333333334
:scan.plan.plan_rows 1250192
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}

```

```

:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
}
:resno 12
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
}
:resno 13
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
}
:resno 14
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
}
:resno 15
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)

```



```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 16
        :location -1
    }
    :resno 16
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    :scan.plan.qual (
    {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 16
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
    )
    :scan.plan.lefttree <>
    :scan.plan.righttree <>
    :scan.plan.initPlan <>
    :scan.plan.extParam (b 0 1)
    :scan.plan.allParam (b 0 1)
    :scan.scanreldid 1
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0 1)
    :plan.allParam (b 0 1)
    :aggstrategy 0
    :aggsplit 6
    :numCols 0
    :grpColIdx ()
    :grpOperators ()
    :grpCollations ()
    :numGroups 1
    :transitionSpace 0
    :aggParams (b)
    :groupingSets <>
    :chain <>
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0)
    :plan.allParam (b 0)
    :num_workers 2
    :rescan_param 1
    :single_copy false
    :invisible false
    :initParam (b 0)
    }
    :plan.righttree <>
    :plan.initPlan (
    {SUBPLAN
    :sublinkType 0
    :testexpr <>
    :paramids <>
    :plan_id 1
    :plan_name InitPlan\ 1
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse true
    :parallel_safe true
    :setParam (i 0)
    :parParam <>
    :args <>
    :startup_cost 0
    :per_call_cost 0.8454845484548454
    }
    )
    :plan.extParam (b)
    :plan.allParam (b 0)
    :aggstrategy 0
    :aggsplit 9
    :numCols 0
    :grpColIdx ()
    :grpOperators ()
    :grpCollations ()
    :numGroups 1
    :transitionSpace 0
    :aggParams (b)
    :groupingSets <>
    :chain <>
    }
    :partPruneInfos <>
    :rtable (
    {RANGETBENTRY
    :alias <>
    :eref
    {ALIAS
    :aliasname lineitem
    :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
    }
    :rtekind 0
    :reldid 16566
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGETBENTRY
    :alias
    {ALIAS
    :aliasname s
    :colnames <>
    }
    :eref
    {ALIAS
    :aliasname s
    :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
    }
    :rtekind 0
    :reldid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    )
    :unprunableRelids (b 1 2)
    :permInfos (
    {RTEPERMISSIONINFO
    :reldid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b)
    :insertedCols (b)
    }
    )

```

```

:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 2818
:scan.plan.plan_rows 3333
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{FUNCEXP
:funcid 17005
:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 2
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16566 16541)
:invalidItems (
{PLANINVALIDITEM
:cacheId 47
:hashValue 2210181313
}
)
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Log 1: Planner Input:
DETAIL: {QUERY

```

:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table <>
:rteperminfos <>
:jointree
{FROMEXPR
:fromlist <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{PARAM
:paramkind 0
:paramid 1
:paramtype 701
:paramtypmod -1
:paramcollid 0
:location 0
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 10
:constvalue 4 [ -24 3 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:location 8

```

```

    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 + reason=outer_reference
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
:QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname *RESULT*
        :colnames <>
      }
    :rtekind 8
    :lateral false
    :inFromCl false
    :securityQuals <>
  }
)
:rteperminfos <>
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals <>
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {OEXPR
        :opno 674
        :opfuncid 297
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {PARAM
            :paramkind 0
            :paramid 1
            :paramtype 701
            :paramtypmod -1
            :paramcollid 0
            :location 0
          }
          {CONST
            :consttype 701
            :consttypmod -1
            :constcollid 0
            :constlen 8
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
          }
        )
        :location 8
      }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ({OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({PARAM :paramkind 0 :paramid 1 :paramtype 701 :paramtypmod -1 :paramcollid 0 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({PARAM :paramkind 0 :paramid 1 :paramtype 701 :paramtypmod -1 :paramcollid 0 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1}) |
parallel_safe = true | reason = PROPARALLEL_SAFE

```

WARNING: Log 3: At the starting of Plan creation:

```
DETAIL: {PROJECTIONPATH
:path.pathtype 331
:parent_reids (b 1)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 0
:path.total_cost 0.0125
:path.pathkeys <>
:subpath
{GROUPRESULTPATH
:path.pathtype 331
:parent_reids (b 1)
:path.pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs <>
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 0
:path.total_cost 0.01
:path.pathkeys <>
:quals <>
}
}
:dummypp true
}
```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```
DETAIL: {PLANNEDSTMT
:commandtype 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 25
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 132849.31333333335
:plan.total_cost 132849.32333333336
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 6
:aggno 0
:aggtransno 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 129723.83333333334
:scan.plan.plan_rows 1250192
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varstype 23
:varstype -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varstype 23
:varstype -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
}
```

```

:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16

```

```

        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 16
        :location -1
    }
    :resno 16
    :resname <>
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanrelid 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggssplit 6
:numcols 0
:grpColIdx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rttable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtkind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
}
:rtkind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:unprunableRelids (b 1 2)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 2818
:scan.plan.plan_rows 3333
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{FUNCEXP
:funcid 17005
:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
}
:location -1
}
}
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 2
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 25
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 132849.31333333335
:plan.total_cost 132849.32333333336
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 6
:aggno 0
:aggtransno 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQRSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 129723.83333333334
:scan.plan.plan_rows 1250192
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3

```



```

:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
}

```

```

:resno 16
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paratype 16
:paratypmod -1
:paramcollid 0
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanrelid 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggsplit 6
:numCols 0
:grpColIdx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rttable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:unprunableRelids (b 1 2)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:resultRelations <>
:appendRelations <>
:subplans (
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 2818
:scan.plan.plan_rows 3333
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{FUNCEXPB
:funcid 17005
:funcresulttype 16
:funcreset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
)
:location -1
}
)
)

```

```

    }
  }
  :scan.plan.lefttree <>
  :scan.plan.righttree <>
  :scan.plan.initPlan <>
  :scan.plan.extParam (b)
  :scan.plan.allParam (b)
  :scan.scanrelid 2
}
}
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalidItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

QUERY PLAN

```

Finalize Aggregate  (cost=133850.37..133850.38 rows=1 width=8) (actual time=627.577..628.695 rows=1.00 loops=1)
  Output: count(*)
  Buffers: shared hit=3132 read=101806
  InitPlan 1
    -> Seq Scan on public.supplier s  (cost=0.00..2818.00 rows=3333 width=8) (actual time=2.209..2.209 rows=1.00 loops=1)
      Filter: check_supplier_balance_safe(s.s_acctbal)
      Buffers: shared hit=218
  -> Gather  (cost=133849.31..133849.52 rows=2 width=8) (actual time=627.398..628.686 rows=3.00 loops=1)
    Output: (PARTIAL count(*))
    Workers Planned: 2
    Workers Launched: 2
    Buffers: shared hit=3132 read=101806
    -> Partial Aggregate  (cost=132849.31..132849.32 rows=1 width=8) (actual time=612.479..612.480 rows=1.00 loops=3)
      Output: PARTIAL count(*)
      Buffers: shared hit=2914 read=101806
      Worker 0:  actual time=606.116..606.117 rows=1.00 loops=1
        Buffers: shared hit=974 read=33536
      Worker 1:  actual time=606.385..606.385 rows=1.00 loops=1
        Buffers: shared hit=959 read=33696
      -> Parallel Seq Scan on public.lineitem  (cost=0.00..129723.83 rows=1250192 width=8) (actual time=0.692..529.988 rows=2000405.00 loops=3)
        Output: lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, lineitem.l_linenum, lineitem.l_quantity, lineitem.l_extendedprice, lineitem.l_discount, lineitem.l_tax, lineitem.l_returnflag, lineitem.l_linestatus, lineitem.l_shipdate, lineitem.l_commitdate, lineitem.l_receiptdate, lineitem.l_shipinstruct, lineitem.l_shipmode, lineitem.l_comment
        Filter: (InitPlan 1).col1
        Buffers: shared hit=2914 read=101806
        Worker 0:  actual time=0.338..525.557 rows=1977814.00 loops=1
          Buffers: shared hit=974 read=33536
        Worker 1:  actual time=0.448..524.799 rows=1985993.00 loops=1
          Buffers: shared hit=959 read=33696
  Planning:
    Buffers: shared hit=27
  Planning Time: 5.411 ms
  Execution Time: 637.458 ms
(31 rows)

tpch=>

```